

413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 2

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RECALL

The conceptual hierarchy in probability theory

$$\text{random experiment} \longrightarrow \Omega \longrightarrow A \subseteq \Omega \longrightarrow P(A).$$

The sample space describes *what can happen*, an event specifies *what we care about*, and the probability measure quantifies *how likely that event is*.

1 Set Theory

Set theory provides a natural language for describing concepts in probability and helps us to visualize a random experiment in terms of collections of possible outcomes and to express events as regions within the space of all possibilities. This viewpoint is useful because many operations on events correspond directly to familiar set operations. For example, intersections represent events occurring together, unions represent at least one of several events occurring, and complements represent an event not occurring.

More concretely, for three coin flips, the sample space

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad |\Omega| = 8.$$

If the event

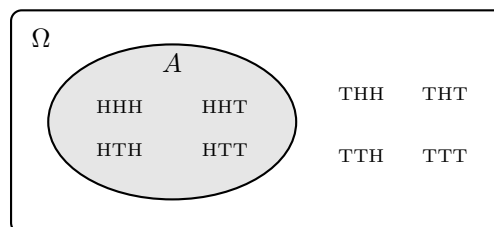
$$A = \{\text{first flip is H}\},$$

then

$$A = \{HHH, HHT, HTH, HTT\}.$$

Since, all outcomes are equally likely, and the Ω is finite, we can use our lazy definition of probability, we get

$$P(A) = \frac{|A|}{|\Omega|} = \frac{4}{8} = \frac{1}{2}.$$



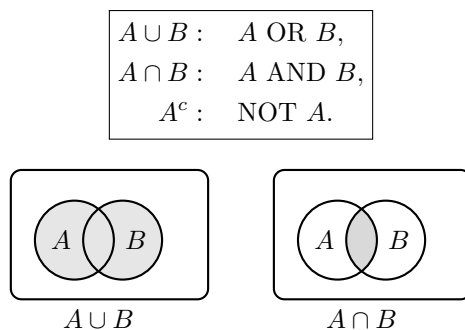
1.1 More than one event

In many probability problems, we are interested not only in a single event, but in how several events are related. For example, we may want to ask whether two events occur together, whether at least one of them occurs, or whether one occurs while another does not.

For example, consider three coin flips, and define

$$A = \{\text{first flip is H}\}, \quad B = \{\text{second flip is T}\}.$$

The language of set theory allows us to translate such logical statements directly into mathematical operations on events. In particular, three fundamental set operations arise naturally:



The union is more inclusive, while the intersection is more restrictive:

$$A \cap B = \{HTH, HTT\}, \quad A \cup B = \{HHH, HHT, HTH, HTT, TTH, TTT\}.$$

$$P(A \cap B) = \frac{|A \cap B|}{|\Omega|} = \frac{2}{8} = \frac{1}{4}, \quad P(A \cup B) = \frac{|A \cup B|}{|\Omega|} = \frac{6}{8} = \frac{3}{4}.$$

1.2 More Probability Rules

The probability axioms allow us to derive several useful rules that we will use throughout the course. In particular,

$$P(A^c) = 1 - P(A), \quad P(\emptyset) = 0.$$

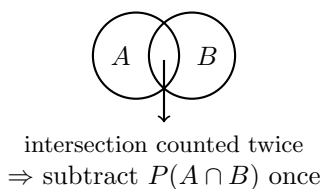
If $A \subseteq B$, then every outcome in A is also contained in B , and hence

$$P(A) \leq P(B).$$

For two arbitrary events A and B , the probability of their union is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The subtraction is necessary because the outcomes in $A \cap B$ are counted once in $P(A)$ and once again in $P(B)$.



It is important to note that, these rules do not require all outcomes to be equally likely. This is important in applications: for example, a coin may be biased, so that

$$P(H) = p, \quad P(T) = 1 - p,$$

with $p \neq \frac{1}{2}$.

1.3 Conditional Probability

Probability provides a mathematical framework for reasoning under uncertainty. In real-world problems, however, our information is rarely fixed: we perform an experiment, observe an outcome, collect new data, or learn that a particular event has occurred. Once this new information becomes available, the probabilities we assign to other events should be updated accordingly. This leads to one of the central questions in probability and statistics:

How should we update our probabilities when we receive new information or evidence?

Conditional probability provides the conceptual framework for answering this question. Suppose we initially have some uncertainty about an event A . We then learn that another event B has occurred. We write

$$P(A \mid B)$$

for the probability of A **given** that B occurred. For events A and B with $P(B) > 0$, the **conditional probability of A given B** is defined by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Conditioning on B means that we restrict our attention to the part of the sample space in which B occurred. Within this restricted sample space, the part corresponding to A is precisely the event space

$$A \cap B.$$

The division by $P(B)$ simply **renormalizes** the probabilities so that the total probability of the new sample space B is one:

$$P(B \mid B) = \frac{P(B)}{P(B)} = 1.$$

1.4 The Multiplication Rule

Starting from the definition of conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$

we immediately obtain

$$P(A \cap B) = P(B)P(A \mid B).$$

Similarly,

$$P(A \cap B) = P(A)P(B \mid A).$$

Therefore,

$$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B).$$

1.5 Independence

So far, we have described how probabilities of events are combined. Another important question is how the occurrence of one event affects what we know about another. We have used the word “independent” informally to describe events that do not influence one another. We now give this idea a precise mathematical definition.

Two events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

The intuition is that learning that one event occurred does not change the probability of the other. Equivalently, whenever the corresponding conditional probabilities are defined,

$$P(A \mid B) = P(A), \quad P(B \mid A) = P(B).$$

Thus, independence means that observing one event provides no additional information about the occurrence of the other. This follows naturally from the multiplication rule,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

1.6 Independent vs. Disjoint

Independence should **not** be confused with disjointness. These two ideas describe very different relationships between events. If A and B are **disjoint**, then they cannot occur together:

$$A \cap B = \emptyset,$$

and therefore

$$P(A \cap B) = 0.$$

Hence, if we learn that A occurred, we immediately know that B did *not* occur. Disjoint events therefore provide strong information about one another. By contrast, if A and B are **independent**, learning that A occurred does not change our probability for B .

Disjoint	: A occurs $\implies B$ cannot occur,
Independent	: A occurs $\implies$ the probability of B does not change.

In fact, if A and B were both disjoint and independent, then

$$0 = P(A \cap B) = P(A)P(B).$$

Therefore,

$$P(A) = 0 \quad \text{or} \quad P(B) = 0.$$

Thus, two events with positive probability cannot be both disjoint and independent. In fact, if A and B are disjoint and both have positive probability, then they cannot be independent.

1.7 Bayes' Rule

As discussed in the previous sections, conditional probability allows us to ask how the probability of an event changes once new information has been observed. Suppose that A is an event of interest and B represents some observed evidence. We would like to relate the probability of A *after observing* B , namely $P(A | B)$, to quantities that may be easier to specify or compute. Recall the multiplication rule:

$$P(B)P(A | B) = P(A)P(B | A).$$

Therefore, provided $P(B) > 0$,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

This is **Bayes' rule**. It provides a precise mathematical form for updating our probability of an event A after observing the evidence for occurrence of B :

$$\underbrace{P(A | B)}_{\text{posterior}} = \frac{\underbrace{P(B | A)}_{\text{likelihood}} \underbrace{P(A)}_{\text{prior}}}{\underbrace{P(B)}_{\text{evidence}}}.$$

The terms have natural interpretations:

- $P(A)$ is the **prior probability**: our probability of A before observing B .
- $P(B | A)$ is the **likelihood**: how compatible the observed evidence B is with A .
- $P(A | B)$ is the **posterior probability**: our updated probability of A after observing B .
- $P(B)$ is the probability of observing the evidence itself and acts as a normalization factor.

Thus, Bayes' rule can be summarized schematically as

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

The evidence $P(B)$ ensures that the updated probabilities are properly normalized. Although Bayes' rule follows from only a few lines of probability algebra, the idea it captures is profound: *new information changes the probability we assign to uncertain possibilities*. This principle lies at the heart of statistical inference and appears throughout modern machine learning. For example, in text-to-image generation, a model transforms a broad distribution over possible images into the conditional distribution

$$p(\text{image} \mid \text{text prompt}),$$

so that a prompt such as “a red fox walking through snow” dramatically changes which images are considered probable. The same idea of conditioning on observed information also underlies inverse problems in science and engineering, where measurements are used to infer unknown physical quantities.

2 Random Variables and Probability Distributions

A random experiment produces an outcome ω in a sample space Ω . The elements of Ω need not be numerical: they may be sequences of coin tosses, configurations of particles, patient outcomes, or trajectories of a stochastic process. To analyze such experiments quantitatively, we associate numerical values with their outcomes.

A **random variable** is a function

$$X : \Omega \longrightarrow \mathbb{R},$$

which assigns a real number to each possible outcome of the experiment. The randomness comes from the experiment, not from the function X itself. Once the outcome ω is known, the value $X(\omega)$ is completely determined. What is uncertain before the experiment is which outcome will occur, and therefore which value of X will be observed.

By mapping outcomes to numbers, random variables allow us to study uncertainty using familiar mathematical quantities such as probabilities, expectations, variances, and distributions.

2.1 Example: Number of Heads

Consider four independent flips of a fair coin. The sample space contains

$$|\Omega| = 2^4 = 16$$

possible outcomes. For example,

$$\omega = HTTH$$

is one possible outcome. A natural random variable is the **number of heads**:

$$X(\omega) = \text{number of heads in } \omega.$$

For this outcome, $X(HTTH) = 2$. Although there are 16 possible outcomes in Ω , the random variable can take only five values:

$$X \in \{0, 1, 2, 3, 4\}.$$

For example,

$$X(TTTT) = 0, \quad X(HTTH) = 2, \quad X(HHHH) = 4.$$

Many different outcomes can therefore produce the same value of the random variable.

Outcomes $\omega \in \Omega$	Random variable $X(\omega)$
$TTTT$	0
$HTTT, THTT, TTHT, TTTH$	1
$HHTT, HTHT, HTTH, THHT, THTH, TTHH$	2
$HHHT, HHTH, HTHH, THHH$	3
$HHHH$	4

A random variable converts outcomes of a random experiment into numerical values:

$$\underbrace{\omega \in \Omega}_{\text{outcome}} \xrightarrow{X} \underbrace{x = X(\omega)}_{\text{numerical value}}.$$

Different outcomes ω may be mapped to the same value x .

2.2 Probability Distribution

Once a random variable X has been defined, the next natural question is:

How likely is each possible value of the random variable?

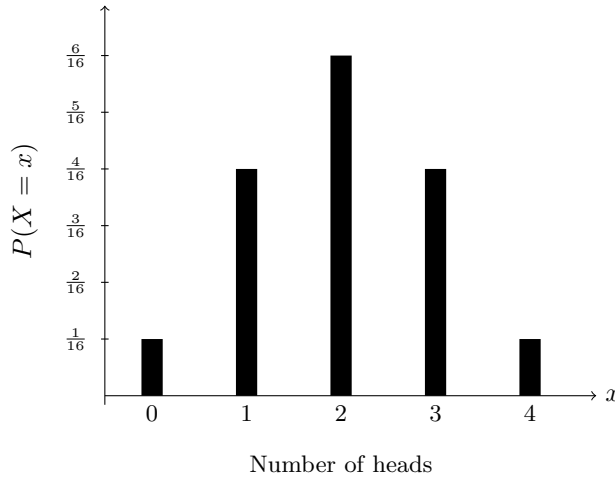
For the four-flip coin example,

x	0	1	2	3	4
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

This collection of probabilities is called the **probability distribution** of X . For example,

$$P(X = 2) = \frac{6}{16},$$

because six of the sixteen possible sequences contain exactly two heads.



Thus, rather than keeping track of all 16 possible sequences, we can summarize the experiment by the distribution of the random variable X .

2.3 Discrete and Continuous Random Variables

Random variables are broadly classified into two types.

Discrete:	X takes a finite or countable collection of values,
Continuous:	X takes values over a continuum.

Examples of *discrete* random variables include the number of heads in repeated coin flips, the number of customers arriving at a store, or the number of defective items in a batch. Random variables can also be *continuous*. Examples of continuous random variables include height, temperature, or the velocity of a molecule.

For a discrete random variable, probabilities are obtained by summing probabilities associated with individual values. For a continuous random variable with probability density $p(x)$, probabilities over intervals are obtained by integration:

$$P(a < X < b) = \int_a^b p(x) dx.$$

So, the probability of observing any one exact value is typically zero.

$$P(X = x) = 0$$

for any specific value of x .

2.4 Normalization

Whenever we introduce a probability distribution, we should check

$$\sum_x P(X = x) = 1$$

for a discrete random variable, or

$$\int p(x) dx = 1$$

for a continuous random variable.

2.5 Mean

If X is a discrete random variable with probability distribution $P(X = x)$, and the function $f(X) = X$, then we can define the expectation as,

$$\mathbb{E}[X] = \sum_x x P(X = x).$$

with the analogous definition

$$\mathbb{E}[X] = \int x p(x) dx$$

for continuous random variables. The mean tells us where the distribution is centered, but it does not tell us how concentrated the distribution is around that value. More generally, we can define the mean/expectation for any function f

$$\mathbb{E}[f(X)] = \sum_x f(x) P(X = x).$$

for X being discrete. If X is continuous with probability density $p(x)$,

$$\mathbb{E}[f(X)] = \int f(x) p(x) dx.$$

Properties of Expectation

1. Scaling by a constant.

$$\begin{aligned} \mathbb{E}[aX] &= \sum_x (ax) P(X = x) \\ &= a \sum_x x P(X = x) = \boxed{a \mathbb{E}[X]}. \end{aligned}$$

2. Adding a constant.

Since $\sum_x P(X = x) = 1$,

$$\begin{aligned} \mathbb{E}[X + c] &= \sum_x (x + c) P(X = x) \\ &= \sum_x x P(X = x) + c \sum_x P(X = x) \\ &= \boxed{\mathbb{E}[X] + c}. \end{aligned}$$

3. Linearity of expectation. For *any* random variables X, Y ,

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_{x,y} (x + y)P(X = x, Y = y) \\ &= \sum_x xP(X = x) + \sum_y yP(Y = y) \\ &= \boxed{\mathbb{E}[X] + \mathbb{E}[Y]}.\end{aligned}$$

Importantly, **independence is not required**.

4. Products of independent random variables. If X and Y are independent, $P(X = x, Y = y) = P(X = x)P(Y = y)$, and therefore

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{x,y} xyP(X = x, Y = y) \\ &= \left(\sum_x xP(X = x) \right) \left(\sum_y yP(Y = y) \right) \\ &= \boxed{\mathbb{E}[X]\mathbb{E}[Y]}.\end{aligned}$$

2.6 Variance

Two random variables can have the same mean but very different levels of fluctuation. We therefore define the **variance** of a random variable X as

$$\boxed{\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]}.$$

The quantity inside the brackets

$$X - \mathbb{E}[X]$$

measures the deviation from the mean. Squaring ensures that positive and negative deviations do not cancel. Expanding the square,

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2,\end{aligned}$$

and hence

$$\boxed{\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2}.$$

The **standard deviation** is defined as

$$\boxed{\sigma_X = \sqrt{\text{Var}(X)}}.$$

Unlike the variance, the standard deviation has the same physical units as the random variable itself.

Properties of Variance

1. Scaling.

$$\begin{aligned}\text{Var}(aX) &= \mathbb{E}[(aX - \mathbb{E}[aX])^2] \\ &= \mathbb{E}[a^2(X - \mathbb{E}[X])^2] \\ &= \boxed{a^2 \text{Var}(X)}.\end{aligned}$$

2. Adding a constant.

$$\begin{aligned}\text{Var}(X + c) &= \mathbb{E}[(X + c - \mathbb{E}[X + c])^2] \\ &= \mathbb{E}[(X + c - \mathbb{E}[X] - c)^2] \\ &= \boxed{\text{Var}(X)}.\end{aligned}$$

Adding a constant shifts the distribution but does not change its spread.

3. Variance of a sum. Define the covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Then

$$\begin{aligned}\text{Var}(X + Y) &= \mathbb{E}\left[\left((X - \mathbb{E}[X]) + (Y - \mathbb{E}[Y])\right)^2\right] \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).\end{aligned}$$

Hence

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If X and Y are independent,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \implies \text{Cov}(X, Y) = 0,$$

so

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad \text{always,}$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are independent.}$$

Expectation is **linear**; variance is generally **not**.

3 Case study: Binomial Distribution

We now apply the concepts of **normalization, expectation, and variance** to one of the most fundamental probability distributions: the **binomial distribution**. The coin-flip experiment is a special case of a more general setting.

A **Bernoulli trial** is a random experiment with two possible outcomes, which we label

$$\{\text{success}\} \quad \text{or} \quad \{\text{failure}\},$$

with

$$P(\text{success}) = p, \quad P(\text{failure}) = q = 1 - p.$$

Now perform N independent Bernoulli trials and define

$$H = \text{number of successes among the } N \text{ trials.}$$

The random variable H can take the values

$$H \in \{0, 1, \dots, N\}.$$

We now ask:

What is the probability of observing exactly h successes?

Consider first one particular sequence containing exactly h successes and $N - h$ failures. By independence, the probability of this particular sequence is

$$p^h q^{N-h}.$$

But there are many different sequences containing exactly h successes. The number of ways to choose which h of the N trials are successes is

$$\binom{N}{h}.$$

Therefore,

$$P(H = h) = \binom{N}{h} p^h q^{N-h}, \quad h = 0, 1, \dots, N.$$

We say that H has a **binomial distribution** and write

$$H \sim \text{Binomial}(N, p).$$

The structure of the binomial probability is useful to remember:

$$\underbrace{\binom{N}{h}}_{\text{number of arrangements}} \times \underbrace{p^h q^{N-h}}_{\text{probability of one arrangement}}.$$

For a fair coin, $p = \frac{1}{2}, q = \frac{1}{2}$, so

$$P(H = h) = \binom{N}{h} \frac{1}{2^N}.$$

For $N = 4$, this gives

$$P(H = h) = \binom{4}{h} \frac{1}{16}, \quad h = 0, \dots, 4,$$

which reproduces the probability distribution obtained in section 2.2.

3.1 Normalization of the Binomial Distribution

A probability distribution must assign total probability 1:

$$\sum_{h=0}^N P(H = h) = 1.$$

For the binomial distribution,

$$P(H = h) = \binom{N}{h} p^h q^{N-h}, \quad q = 1 - p.$$

Therefore,

$$\sum_{h=0}^N P(H = h) = \sum_{h=0}^N \binom{N}{h} p^h q^{N-h}.$$

To understand this sum, recall the **binomial theorem** (not covered in this course). For example,

$$(p + q)^2 = q^2 + 2pq + p^2,$$

and

$$(p + q)^3 = q^3 + 3pq^2 + 3p^2q + p^3.$$

Notice that the coefficients

$$1, 3, 3, 1$$

are exactly the binomial coefficients

$$\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}.$$

More generally,

$$(p + q)^N = \sum_{h=0}^N \binom{N}{h} p^h q^{N-h}.$$

Thus,

$$\sum_{h=0}^N P(H = h) = (p + q)^N.$$

But since

$$q = 1 - p,$$

we have

$$p + q = 1.$$

Therefore,

$$\sum_{h=0}^N P(H = h) = (p + q)^N = 1^N = 1.$$

Hence the binomial distribution is properly normalized.

3.2 Expectation and Variance of the Binomial Distribution

The binomial distribution tells us the probability of every possible number of successes,

$$P(H = h) = \binom{N}{h} p^h (1 - p)^{N-h}.$$

Knowing the full distribution is useful, but in many applications we would like to summarize it using only a few characteristic quantities. Two of the most important are:

- the **mean**, which describes the typical or average number of successes;
- the **variance**, which measures how strongly the number of successes fluctuates around that mean.

Thus, our goal is to answer two questions:

Where is the distribution centered? and How wide is it?

3.2.1 The Mean

For a discrete random variable H , the expectation is defined by

$$\mathbb{E}[H] = \sum_{h=0}^N h P(H = h).$$

For the binomial distribution, this gives

$$\mathbb{E}[H] = \sum_{h=0}^N h \binom{N}{h} p^h q^{N-h}, \quad q = 1 - p.$$

This sum can be evaluated directly, but there is a convenient systematic tool that will also allow us to compute higher moments. Define the **probability generating function**

$$G(z) = \mathbb{E}[z^H] = \sum_{h=0}^N P(H = h) z^h.$$

For a binomial random variable,

$$\begin{aligned} G(z) &= \sum_{h=0}^N \underbrace{\binom{N}{h} p^h q^{N-h}}_{P(H=h)} z^h \\ &= \sum_{h=0}^N \binom{N}{h} (pz)^h q^{N-h}. \end{aligned}$$

Using the binomial theorem again, we can reduce the above expression to

$$G(z) = (q + pz)^N.$$

As a first check,

$$G(1) = \sum_{h=0}^N P(H = h) = (q + p)^N = 1^N = 1,$$

so normalization is recovered automatically. Now differentiate the definition of G :

$$G'(z) = \sum_{h=0}^N h P(H = h) z^{h-1}.$$

Evaluating at $z = 1$ gives

$$G'(1) = \sum_{h=0}^N h P(H = h) = \mathbb{E}[H].$$

On the other hand, differentiating the closed-form expression gives

$$G'(z) = Np(q + pz)^{N-1}.$$

Therefore,

$$\boxed{\mathbb{E}[H] = G'(1) = Np(q + p)^{N-1} = Np.}$$

Thus,

$$\boxed{\text{mean number of successes} = \text{number of trials} \times \text{probability of success.}}$$

For example, if a fair coin is flipped N times,

$$p = \frac{1}{2} \quad \implies \quad \mathbb{E}[H] = \frac{N}{2}.$$

Notice that $\mathbb{E}[H]$ need not itself be a possible outcome. For example, if $N = 5$ and $p = 1/2$, then

$$\mathbb{E}[H] = 2.5,$$

even though the observed number of heads must be an integer.

3.2.2 Variance of a Binomial Random Variable

To compute $\text{Var}(H)$, we need $\mathbb{E}[H^2]$. Differentiate the generating function twice:

$$G''(z) = \sum_{h=0}^N h(h-1)P(H = h)z^{h-2}.$$

Therefore,

$$\boxed{G''(1) = \mathbb{E}[H(H-1)].}$$

From

$$G(z) = (q + pz)^N,$$

we obtain

$$G''(z) = N(N-1)p^2(q + pz)^{N-2},$$

and hence

$$\mathbb{E}[H(H-1)] = G''(1) = N(N-1)p^2.$$

Now use the relation

$$H^2 = H(H-1) + H.$$

Taking expectations,

$$\begin{aligned} \mathbb{E}[H^2] &= \mathbb{E}[H(H-1)] + \mathbb{E}[H] \\ &= N(N-1)p^2 + Np. \end{aligned}$$

Substituting the expressions in the variance definition,

$$\begin{aligned} \text{Var}(H) &= \mathbb{E}[H^2] - \mathbb{E}[H]^2 \\ &= N(N-1)p^2 + Np - N^2p^2 \\ &= Np - Np^2 \\ &= \boxed{Np(1-p)}. \end{aligned}$$

Thus,

$$\boxed{\sigma_H = \sqrt{Np(1-p)}}.$$

How Do Fluctuations Scale with N ?

The mean number of successes grows linearly with the number of trials:

$$\mathbb{E}[H] = Np \sim N.$$

The variance also grows linearly,

$$\text{Var}(H) = Np(1 - p) \sim N,$$

but the standard deviation grows only as

$$\sigma_H \sim \sqrt{N}.$$

This distinction is important. As N increases, the *absolute* fluctuations in the number of successes become larger. However, they grow much more slowly than the mean. To measure the size of the fluctuations relative to the typical value, consider

$$\frac{\sigma_H}{\mathbb{E}[H]} = \frac{\sqrt{Np(1-p)}}{Np}.$$

Simplifying,

$$\frac{\sigma_H}{\mathbb{E}[H]} = \sqrt{\frac{1-p}{Np}} \sim \frac{1}{\sqrt{N}}, \quad 0 < p < 1 \text{ fixed.}$$

Thus, although the number of successes fluctuates more in absolute terms as N increases, those fluctuations become progressively smaller as a fraction of the total.

For N independent Bernoulli trials,

$$\mathbb{E}[H] = Np, \quad \text{Var}(H) = Np(1-p), \quad \sigma_H = \sqrt{Np(1-p)}.$$

The relative fluctuations satisfy

$$\frac{\sigma_H}{\mathbb{E}[H]} \sim \frac{1}{\sqrt{N}}.$$

This is one of the first indications of a fundamental statistical principle: averages and proportions become increasingly stable as the number of independent observations grows.